CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M.Tech Examination (CE)
December 2012

CE713: Data Mining & Business Intelligence

Date:13/12/2012 ,Thursday | **Time: 10:00 am to 01:00 pm** | **Maximum Marks:70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1

(a) Why on-line analytical processing is not performed directly on databases, instead of spending additional time and resources to construct a separate data warehouse. **2**

(b) Write down the Bayes' Theorem and explain Naive Bayes Classification in brief. **2**

(c) What is concept hierarchy? How concept hierarchies are useful in rollup, drilldown, slice and dice OLAP operations? **3**

Q-2

(a) Explain the architecture of a typical data mining system with their major components. **6**

(b) In real-world data, tuples with missing values for some attributes are a common occurrence. Describe various methods for handling this problem. **4**

(c) Describe attribute oriented induction with example. **4**

OR

Q-2

(a) Define Data cube measure. Differentiate between distributive, algebraic and holistic measure with an example. **6**

(b) Briefly outline the k-means partitioning algorithm for clustering. **4**

(c) Give the difference between OLTP and OLAP. **4**

Q-3

(a) A data warehouse can be modeled by either a star schema or a snowflake schema. Briefly describe the similarities and the differences of the two models, and then analyze their advantages and disadvantages with regard to one another. **6**

(b) How sampling can be used as a data reduction technique? Explain with an example Simple random sample without replacement, Simple random sample with replacement, Cluster sample and Stratified sample. **8**

OR

(b) Explain with the use of a flowchart to summarize the following procedures for attribute subset selection: **8**

(a) stepwise forward selection
(b) stepwise backward elimination
(c) a combination of forward selection and backward elimination
(d) Decision Tree Induction

SECTION-II

Q-4

(a) Define following terms. **4**

1) Supervised Learning
2) Attribute Oriented Induction
3) Unsupervised Learning
4) Frequent Item Set

(b) How is decision tree used for classification? Why are decision tree classifiers so popular? **3**

Q-5

(a) Many variations of the Apriori algorithm have been proposed that focus on improving the efficiency of the original algorithm. Explain following variation in brief. **6**

1. Hash-based technique
2. Transaction reduction
3. Partitioning

(b) Write short note on Data Reduction Techniques. **4**

(c) Explain Performance Issue in Data Mining. **4**

OR

Q-5

(a) Explain requirements of clustering in data mining **6**

(b) Explain two basic measures for content-based text retrieval. **4**

(c) Explain the criteria by which Classification and prediction methods can be compared and evaluated. **4**

Q-6 Attempt Any Two. **14**

(a) Consider the following data (in increasing order) for the attribute *age*: 13, 15, 16, 16, 19, 20,20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

(1) Use min-max normalization to transform the value 35 for *age* onto the range [0.0, 1.0].

(2) Use z-score normalization to transform the value 35 for *age*, where the standard deviation of *age* is 12.94 years.

(3) Use normalization by decimal scaling to transform the value 35 for *age*.

(b) Suppose that a data warehouse consists of the three dimensions *time, doctor*, and *patient*, and the two measures *count* and *charge*, where *charge* is the fee that a doctor charges a patient for a visit.

(a) Enumerate three classes of schemas that are popularly used for modeling data warehouses.

(b) Draw a schema diagram for the above data warehouse using star schema .

(c) Starting with the base cuboid [*day, doctor, patient*], what specific *OLAP operations* should be performed in order to list the total fee collected by each doctor in 2012?

(c) A database has four transactions. Let *minimum support* = 50% and *minimum confidence* = 70%.

TID	Items
T1	A,C,D
T2	B,D
T3	A,B,C,E
T4	B,D,F

Find all frequent itemsets using Apriori and list all *strong* association rules with support and confidence.